

Forecasting Volatility in Pakistan Stock Exchange



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2018



In The Name of Allah The Most Merciful and The Most Beneficent

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*A THESIS SUBMITTED IN THE PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE
DEGREE OF BACHELOR IN STATISTICS*

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2018

Declaration

I Muhammad Shahzeb Ali, declare that this thesis titled, “Forecasting Volatility in Pakistan ”.

- This work was done wholly in candidature for a degree of BS Statistics at this University.
- Where I got help from the published work of others, this is always clearly stated.
- Where I have quoted from the work of others, the source is always mentioned. Except of such quotations, this thesis is entirely my own research work.
- Where the thesis is based on work done by myself jointly with my supervisor, I have made clear exactly what was done by others and what I have suggested.

Signed:

Date:

Dedication

I am feeling great honor and pleasure to dedicate this research work to

My beloved Grandmother (Late)

My beloved mother

And My Uncle

Dr. Mumtaz hussain

Whose endless affection, prayers and wishes have been a great source of comfort for me during my whole education period.

Acknowledgments

First and foremost I praise and acknowledge Allah Almighty, the Lord and creator of the heavens and earth. All respect and gratitude goes to the Holy Prophet Muhammad (Peace be upon him) who enlightens our hearts with the light of Islam and whose way of life has been a continuous guidance for us.

I deem it my utmost pleasure to avail this opportunity and express gratitude and deep sense of obligation to my supervisor **Dr. Ismail Shah** for his valuable and dexterous guidance, scholarly criticism, untiring help, compassionate attitude, kind behavior and moral support. I wish to express my sincere thanks for his great guidance, support, valuable suggestions and motivation for this research. I am greatly indebted to him for helping and guiding me throughout the thesis in every aspect.

I offer my deepest sense of gratitude, profound respect and tribute to all those teacher who have enlightened my statistical knowledge throughout my academic career and guidance throughout research work.

My sincere thanks go to my mother for her love and support throughout my life. I owe my loving thanks to my sisters Shahbano, Shahtaj and Shahrose and my whole family for their great love and support. Without their support and encouragement it would have been impossible for me to complete this work. I would like to thank all my friends and classmates for their cooperation and help especially Shahid khan, Sohail yousafzai, Sir Zulfiqar, Sidra api, Maam Sunila, Memona Nazir .Finally, I would say thanks to all who have been supportive and cooperative to me during my research work.

Muhammad Shahzeb Ali

Abstract

Analysis of time series is used to developed simple models which are able to forecast, interpret and analyze the result concerning its field of application. The actual purpose of this study was to check and model the volatility in Pakistan stock Exchange for the near future. The stock exchange data is highly volatile and there are many factors which effect the daily market. For example political stability or instability, dollar prices, Monday charisma, last working day of week etc. So, it becomes very difficult to forecast these returns due to continuous effect of above and many other factors. In these conditions the univariate models i.e., Autoregressive (AR), Moving Average (MA), Autoregressive integrated moving average (ARIMA) unable to give precise forecast because they are not well good to capture sudden changes in market. We applied these models on our data as well to make their comparison with the volatility models i.e., Autoregressive conditional heteroscedastic (ARCH) , Generalized Autoregressive conditional heteroscedastic (GARCH) and Exponentially Generalized Autoregressive conditional heteroscedastic (EGARCH) models. The data period was 5 years i.e., from January 1st 2013 to 31st December 2017. This time period was selected because in this time there was a political change in Pakistan politics and coincidently oil prices were at their lowest rate in international market, which affect the financial market directly. So I took this data to make government to government comparison also if someone else interested. The result of comparison will helps us to understand political factor, which I mention above. The best model selection is never easy and I apply the following criterias to choose best model among all for forecasting.

- I. MAPE (Mean absolute percentage error)
- II. Root Mean square error (RMSE)

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Chapter 1

Pakistan Stock Exchange and Volatility Analysis

This chapter describes the history and formation of Pakistan Stock Exchange (PSX), Daily stock market returns of different countries, the contribution of PSX in international financial market and different factors effecting daily stock returns. In addition, the objective, significance, and organization of the study is given at the end of this chapter.

1.1 Introduction

The issue pertaining to volatility in stock market have always been the interest of the researcher in modern financial theory as it has a significant impact on the investment. Volatility, apparently an easy and discerning concept, refers to unexpected return due to unexpected events resulting in huge price movement with non-constant variance. Analyzing the foreign stock markets by volume indicates that globally 93 trillion United states (US) dollar amount of trading occurred worldwide in the fiscal year 2017, whereas the share of Pakistan was 27.53 billion US dollar. Therefore, it is not surprising that stable stock returns has served as a catalyst of economic growth for most developed countries while a highly volatile stock market returns is a major problem for many developing countries.

Karachi Stock Exchange (KSE) was a stock index acted as a benchmark to collect, compare and analyze prices of different stock markets until it was replaced by PSX recently. In determining representative companies to compute the index, companies with the highest market capitalization are selected. However, to ensure full market representation, the company with the highest market capitalization from each sector is also included. Nearly a decade ago, the Securities and Exchange Commission of Pakistan (SECP) released its first report on the demutualization of the stock market, the process where ownership and

trading rights at the exchanges are separated from each other. The idea was to usher in an era to privately owned and operated stock markets so that the necessary investment in upgrading the infrastructure of exchanges could be mobilized and superior oversight could restrain the power of the large brokers to manipulate the market and engage in unethical trading practices. The integration of the three exchanges is an essential step in that direction. With three separate stock markets, and with one of them dominant over other two, the search for a private investor was complicated because it was hard to find an interested party for the Lahore and Islamabad exchanges, given their tiny size, and any private investor was wary of acquiring ownership of one exchange in Karachi while the other two remained in government hands.

Pakistan stock exchange (PSX) is the stock exchange of Pakistan with trading floors in Karachi, Islamabad and Lahore. PSX was established on 11 January 2016 after the merger of individual stock exchanges of Karachi, Lahore and Islamabad. PSX origin was laid with the establishment of the Karachi Stock Exchange in 1947, the Lahore Stock Exchange in 1970 and the Islamabad Stock Exchange in 1992. As of February 23, 2018 there are 559 companies listed in PSX and the total market capitalization is 84 billion US dollars. PSX was reclassified as a emerging market by Morgan Stanley Capital International (MSCI) in May 2017, while the Financial Time Stock Exchange (FTSE) classifies PSX as a Secondary Emerging Market.

The investors on the exchange include 1886 foreign and 883 domestic institutional along with about 0.22 million retail investors. There are also about 400 brokerage houses which are members of the PSX as well as 21 asset management companies. It is among the worlds best performing stock markets between 2009 to 2015 by delivering 26% a year. In December 2016, PSX sold 40% strategic shares to a Chinese consortium for 85 million US dollars.

The question of whether stock markets are integrated has gained enormous importance in the empirical literature in the last two decades. According to [Bekaert et al. \(2002\)](#), asset returns in segmented markets are determined by the local risk factors and associated risk premia while for integrated markets global risk factors and global risk premia are relevant. Recent studies are more inclined towards portfolio allocation benefits. The economic motivation behind these studies is to establish whether some markets provide diversification gains. The simplest measure of market integration is provided by the correlation coefficient between the country market indices. For the period February 1993 to January 1996, Smith and Walter (1998) report that the correlation of Pakistans equity market with the

US market is -0.01. In [Harvey \(1995\)](#), the correlation of Pakistans equity market with the MSCI developed market index and the overall world market index is reported to be 0.02 and 0.04, respectively. To investigate integration of equity markets, [Korajczyk \(1996\)](#) used a measure of integration based on pricing errors from an international Arbitrage pricing theory (APT) model. Using monthly data of 52 stocks from Pakistans market for the period 1984-1992, the study reports that pricing errors for Pakistan are relatively smaller but contrary to the intuition, the pricing errors were higher in the post-liberalization period. [Hussain and Saidi \(2000\)](#) used cointegration analysis to investigate about integration of Pakistan equity market with the developed markets of US, United Kingdom (UK) and Japan, among others. Although cointegration was found, which indicates that prices of Pakistan equity market move with these markets, any causal effect of these markets on Pakistan was not detected. This observation led them to conclude that Pakistans equity market has diversification potential. Similarly, [Naeem \(2002\)](#) found cointegration of Pakistan's equity market with the US and UK in the pre-nuclear test period ranging from January 1994 to April 1998 but not for the entire sample including a later period. Using weekly country level index data from November 1987 to November 1999, [Darrat and Zhong \(2002\)](#) analyzed the permanent and transitory impact of US and Japans equity markets on a set of 10 Asian markets including Pakistan. The study found that, like most other countries included in the sample, Pakistan's market appears to be driven permanently by the US market whereas the effect of the Japanese market is transitory. [Lamba \(2005\)](#) used cointegration to test the dynamic relationship of the South Asian markets including Pakistan, India, and Sri Lanka with the developed markets. Using monthly data from July 1997 to February 2003, Lamba found that while the Indian market is cointegrated with the US market, Pakistan market appears to be relatively isolated. Summarizing this evidence it appears that, like many emerging markets, Pakistan is not yet integrated with the developed markets.

Financial market channel is saving the efficient investments to facilitate economic growth and development. However, stock market volatility may be an obstacle in this process especially in an emerging economy where high volatility in prices leads to erosion of capital from the market. As such, what causes high volatility in the stock market is a continued discussion among the market experts and academicians. The financial markets develop an unexpected behavior that may confuse the investors. Stock market in Pakistan is highly volatile as it is very sensitive and reactive to unanticipated shocks and news. It takes no time to affect the market activities. However, at the same time Pakistani stock market

is resilient that recovers soon after shocks. An increased participation is observed of the individual investors in the financial markets over recent time. As a result, their behavior, actions, reactions, and perceptions observed as individual investors to manifest themselves on a much larger scale in the overall stock market. They lead to pricing anomalies and unexplainable movements in stock prices. Behavioral finance seeks to elucidate the unjustifiable stock price volatility.

Volatility is a vital input to determine the overall cost of capital. Stock prices generally demonstrate nonlinear and probably chaotic behavior. However, some observe that stock returns are predictable imperfectly in the short-run but unpredictable in the long-run and statistical distribution can measure the return unevenness. Literature has identified a number of factors that cause stock market volatility. For example, credit policy, inflation, interest rate, corporate earnings, financial leverage, dividend policies, bond prices, and many other macroeconomic, social and political variables that greatly effect the market. In Pakistan, political instability, terrorism and many other foreign influences like International monetary fund (IMF) conditionalities, inconsistent monetary and fiscal policies, cause more volatility in stock returns. Despite the importance of the subject, limited research is available on this topic with regard to Pakistan.

Global financial crises (2007-2008) shocked every investor, all the research work looks useless because no one could predict this sudden collapse. The study of this collapse is beyond the scope of this topic but its urge the researcher to change their way of study, remove the models which are old enough and not compatible of capturing sudden changes, find the new way of studying financial markets and to develop new and stronger models which can model and predict volatility in stock markets more efficiently. Hence forecasting the volatility in stock market is challenging and more demanding issue for investors.

As far as a risk-averse investor concerned, the state of being uncertain is the most important fact in pricing any financial asset. Hence the strong asset prices theories try to model this uncertainty or risk by using the covariance between asset return and the market portfolio. It has been recognized for quite some time that the uncertainty of unpredictable prices, as measured by variance and covariance, is changing through time. Thus, the researchers have started to explicitly model this variation in second or higher order moments. The progress in the area of accurate volatility modeling for in financial market has greatly over elaborated in decision making process for highly risk-averse as opposed to risk-neutral investors.

Variance or standard deviation which is the traditional measure for volatility is uncon-

ditional and does not recognize the patterns in asset volatility such as time-varying and clustering properties. Researchers have therefore introduced various models to explain and predict these patterns in volatility. Engle (1982), is the first who introduce the Autoregressive conditional heteroskedasticity (ARCH) model to measure volatility. The ARCH models the heteroscedasticity by relating the conditional variance of the disturbance term to the linear combination of the squared disturbance in the recent past. Bollerslev (1986), generalizes the ARCH model by modeling the conditional variance to depend on its lagged values as well as squared lagged values of disturbance, and he called this generalized autoregressive conditional heteroskedasticity (GARCH) model. The ARCH/GARCH models has made a significant effect on the development of financial econometrics in the past two decades. The GARCH model is useful for interpreting the volatility-clustering effect, which is an important angle of asset prices stated by Mandelbrot (1963). This model is also helpful for many financial applications, such as measuring volatility of financial market and pricing options, suggested by Engle (2002). Engle (1982) suggests that forecasting exchange rate volatility, Ordinary Least Square (OLS) maintains its optimality properties however, it provide a better estimates when Maximum Likelihood (ML) method is employed. Zivot (2009) provides a detailed empirical analysis of GARCH models for financial time series which emphasis on practical issues associated with model specification, estimation, diagnostics, and forecasting and finds the GARCH model to provide better description of financial data as compared to the OLS. Bala and Asemota (2013) concluded that the GARCH model has dominated the literature on volatility since the early 1980s, because the model allows for preservance of conditional variance by imposing an autoregressive structure on squared errors of the process showing the excellence of the ARCH/GARCH models over other models for forecasting volatility.

Engle (1982) and Bollerslev (1986) suggests various forms of GARCH model for modeling volatility. In the view of that, Hammoudeh and Li (2008) have analyzed sudden change in volatility for five Gulf area stock market and found that GARCH (1,1) accounts for the large shifts in volatility reduces the estimated persistence of the volatility for the Gulf stock markets significantly. In addition to the preceding models, other research work indicate that the behavior of dealers and other market participants can influence fluctuations in stock market, Evans and Lyons (2002). Furthermore, it is most likely that the macroeconomic information and order flow information act in a nonlinear fashion to the volatility problem.

Several techniques are used to forecast the volatility in stock market including univari-

ate time series models and multivariate models, also known as forecasting based models. These models include Autoregressive (AR) model, Moving average (MA) model, Autoregressive integrated moving average (ARIMA), Autoregressive conditional heteroscedasticity (ARCH), Generalize Autoregressive conditional heteroscedasticity (GARCH) models. All these models are extensively used for forecast problems in economics, business, and other related areas of study.

1.2 Objective of the study

The study is conducted by keeping in view the following objectives:

- To compare the performance of univariate models for stock market including those models that can account for volatility in the data
- To account for any exogenous variable if available
- To find out appropriate forecasting model for stock market volatility

1.3 Significance of study

For any business investment decision, stock market volatility forecast is an important input. Moreover, the investor of stock market need stock market volatility forecast for proper opinion regarding speculation and asset. This study will help foreign companies, banks, individuals, investors and investment management firms etc. to forecast volatility in stock market by using efficient forecasting model. There are different methods for prediction and forecasting the data. Time series modeling is one of the best method which deal with the time based data. It includes the working on time (years, days, hours and minutes) series data to evolve hidden vision in decision making. Furthermore, in case of serially correlated data, time series models are generally best to use.

1.4 Organization of study

The study proceeds as follows. Chapter 2 provides the related literature review. Chapter 3 provides description of data and stationarity of the data. chapter 4 consist of different volatility models and their forecasted results, while Chapter 5 gives compile result of all models, discussion of results and also summary of whole thesis and use of study in further directions.

Chapter 2

Literature Review

In statistics, analysis of time series is an important tool which is used to evaluate uniqueness of data to predict future values. Time series forecasting is the process of using a model to generate predictions (forecasts) for future events based on known past events. The goal of forecasting is to project the underlying trend or pattern of the time series into the future to obtain most likely values for the future. Forecasting is the combination of knowledge from the past and future expectations with an estimated model to produce likely outcomes for the future, hence reducing the uncertainty inherent in the decision-making process.

Volatility is a statistical measure of the dispersion of returns for a given security or market index. Volatility in terms of stock market is defined as the sudden changes (jumps) or high fluctuations in index or price within short time interval. It can be measured using the standard deviation or variance between returns from the same security or market index. Generally, the higher the volatility, the riskier the security. Thus, when predicting prices or index, one should pay more attention to incorporate the component of volatility in the model in order to have more accurate prediction.

Stock market is the largest financial market of any country and hence, is very important to forecast the volatility in stock market as it has diversified reaction to our daily life. Without forecasting volatility in stock market, business can suffer heavy loss that can result in increase or decrease of assets prices. Therefore, a variety of models have been proposed for this task in the past. For example, [Franses and Van Dijk \(1996\)](#) study the performance of the GARCH model and two of its non-linear modifications to forecast weekly stock market volatility. Those models were Quadratic GARCH and the Glosten Jagannathan and Runkle (GJR) models which have been proposed to describe, for example, the often observed negative skewed behaviour in stock market indices. They found that the QGARCH model is best when the estimation sample does not contain extreme observations such as the 1987

stock market crash and that the GJR model cannot be recommended for forecasting.

The temporal behavior of the stock market has received increasing interest gradually. The upward trend in volatility during 90's has been attributed to the major reason for the decline in equity prices ([Pindyck, 1984](#)). Pindyck argues that due to "the variance of the firms", real gross marginal return on capital has increased significantly since 1965, and this has increased the relative risk of investor's net real returns from holding stocks, resulting in market decline. [Chou \(1988\)](#) focused on volatility persistence and the changing risk premium in the stock market using the GARCH estimation technique. He computed an estimate of the index of relative risk aversion to be 4.5 that confirm the existence of changing equity in the US during 1962 to 1985. The persistence of shocks to the stock return volatility was so high that the data cannot reject a non-stationary volatility process specification. The results of his paper were consistent with Pindyck's hypothesis about the unforeseen rise in the investment uncertainty during 1974 that causes the market to plunge.

[Dimson and Marsh \(1990\)](#) suggest that accurate forecast can be obtained by simple methods, and the use of exponential smoothing and simple regression models were recommended, however, they did not subject ARCH models to examination. [Baillie and Myers \(1991\)](#) applied Bi-variate GARCH model on cash and future in commodities prices. The optimal hedge ratio (OHR) is calculated as a ratio of the conditional covariance between cash and futures to the conditional variance of future.

The findings of clustered volatility and ARCH effects is ubiquitous in financial data. [Lux and Marchesi \(2000\)](#) presents a possible explanation for this phenomenon within a multi-agent framework of speculative activity. In their study both, chartist and fundamentalist, approaches were considered with agents switching between behavioral variants according to observed differences in pay-offs. The clustered volatility was clarified as the outcome of market being the focus to rare short-term volatility.

The discussion about comparison of performance of linear and non-linear models is always an interested topic for researchers. [Gokcan \(2000\)](#) found that ARCH and GARCH models are substantially used for modeling volatility of time series data. It is proven by many studies that if variables are significantly skewed, linear versions of these models are not sufficient for both, explaining the past volatility and forecasting future volatility. In their study they compare the linear GARCH (1,1) and non-linear E-GARCH model by using the monthly stock market returns of seven emerging countries from February 1988 to December 1996. They found that for emerging stock market GARCH (1,1) model performs better

than EGARCH model, even if stock market return series display skewed distributions.

[Yu \(2002\)](#) evaluate the performance of nine alternative models for predicting stock price volatility using New Zealand data. The competing models contain both simple models such as the random walk, smoothing models, and complex models such as ARCH-type models and a stochastic volatility model. Four different measures were used to evaluate the forecasting accuracy. They concluded that the stochastic volatility model provides best performance among all the candidates whereas ARCH-type models can perform well or badly depending on the form chosen. Furthermore, the performance of the GARCH (3,2) model, the best model within the ARCH family, is sensitive to the choice of assessment measures, and the regression and exponentially weighted moving average models do not perform well according to any assessment measure, in contrast, to the result found for various markets.

Several studies investigated the performance of GARCH models in explaining the volatility of emerging stock markets. [Mangani \(2005\)](#) examined expected return and volatility of stock market using GARCH models. The results show that the effects of shocks on volatility are symmetric and that volatility is not a commonly priced factor.

Groen (2005) tests the long run relationship between monetary fundamental and returns and finds that there exist a strong relation between the monetary fundamental and the long run returns of the countries involved. The research further tests the out of sample performance of the model with the random walk and found that the cointegrated VAR model based forecasts best performs than random walk, especially for forecasting horizon of 2 to 4 years. [Beltratti and Morana \(2006\)](#), examined the association among macroeconomic and capital market volatility, using Standard and Poor's 500 index (S&P 500) data for the period 1970-2001. They found confirmation of both long memory and structural change in the volatility process. [Liu and Shrestha \(2008\)](#) observed the relation between the macro factors and share prices in the China stock market and found that strong relationship between these variables exist.

[Ahmed and Suliman \(2011\)](#) study Khartoum Stock Exchange (KSE) for the period from Jan 2006 to Nov 2010. Their results show high level of persistence in the stock market which indicates low-risk aversion. Sabrina Khanniche (2015) studied hedge fund risk and their volatility using symmetric (GARCH) and asymmetric (EGARCH and TGARCH) models. They forecasted one-day ahead value at risk, and concluded that GARCH performs better than the rest.

[Adam and Zhu \(2015\)](#) found that consumption based asset pricing models with time sepa-

rable preferences generate realistic amounts of stock price volatility if one allows for small deviations from rational expectations. Rational investors with subjective beliefs about price behavior optimally learn from past price observations. This imparts momentary and mean revision into stock prices. The model quantitatively accounts for the volatility of returns, the volatility and persistence of the price-dividend ratio, and the predictability of long horizon returns. It passes a formal statistical test for the overall fit of a set of moments provided one excludes the equity premium.

[Kang et al. \(2015\)](#) examine the impact of oil price structural shocks on the U.S stock market return and stock market volatility. They use daily data on return and volatility and construct the covariance of return and volatility at monthly frequency. The measures of daily volatility are realized-volatility at high frequency, conditional volatility recovered from a stochastic volatility model, and implied-volatility deduced from options prices. It is observed that Positive shocks on aggregate demand and oil market specific demand are associated with negative effects on the covariance of return and volatility. Oil supply disruptions are associated with positive effects on the covariance of return and volatility. The spillover index between the oil price structural shocks and covariance of stock return and volatility is large and highly statistically significant.

[Asif and Aziz \(2016\)](#) investigate the cluster volatility of return distribution in the PSX. GARCH model for characterizing financial market volatility is used in their work. They found that GARCH (1,1) is most appropriate model among ARCH family models for capturing stock market volatility. The outcome of their study indicates that PSX is weak-form efficient and explains cluster volatility and leptokurtic distribution. As Pakistan is directly effected by terrorism since a decade that resulted devaluation of our economy as the economy and stock markets are directly linked with terrorism. To investigate this, [Balcilar et al. \(2018\)](#) used nonparametric causality in quantiles test to study the effect of terrorism attacks on stock market returns and volatility in G7 countries. They also used the test to study the international repercussions of terrorism attacks. The results showed that terror attacks often have significant effects on returns, whereas the effect on volatility is significant only for Japan and UK for several quantiles above the median.

[Dimpfl and Jank \(2016\)](#) study the dynamics of stock market volatility and retail investors attention to the stock market. The latter is measured by Internet search queries related to the leading stock market index. They found a strong co-movement of the Dow Jones realised volatility and the volume of search queries for its name. Furthermore, including search queries in autoregressive models of realised volatility improves volatility forecasts

for in-sample, for out-of-sample, for different forecasting horizons, and in particular in high volatility phases.

Ahmed(2017) investigated the use of ARCH family models for forecasting volatility of four regional emerging markets i.e., Karachi Stock Exchange (KSE-100), SP Bombay Stock Exchange Sensitive Index (BSE-SENSEX), Dhaka stock exchange (DSE 20) and Shanghai Stock Exchange (SSE). The ARCH, GARCH, EGARCH, TGARCH and PARCH models were used and the best model was selected on the basis of Akaike information criteria (AIC) and Schwartz information criteria (SIC) over the sample period ranging from January 1996 to December 2015. Empirical evidence suggested on the basis of AIC that TGARCH outperformed other models in case of BSE SENSEX, DSE 20 and SSE index. Whereas, on the basis of SIC, GARCH is the best performing model for BSE SENSEX and SSE, and PARCH and EGRACH for KSE 100 and DSE 20 respectively.

Chapter 3

Time series and Predictive models

This chapter gives an introduction to the features of time series and provide a comprehensive review of the predictive models used in this study. Different methods for a model selection are also discussed in this chapter.

3.1 Time series and stationarity

A time series is a set of observations generated sequentially in time. The series may be represented as $Y_1, Y_2, \dots, Y_T$, where $t = 1, 2, \dots, T$ refers to time period or instants on which the random variable Y is observed. Y_t is said to be deterministic if it is completely determined by any mathematical formula or equation. If any probability distribution is used for determining the future values, it will be called as stochastic process. Stochastic process may be stationary or non-stationary. A special class of stochastic process is called stationary if it has constant mean, variance and no autocorrelation over the time. In other words, random process is said to be stationary if it's mean and variance has no systematic change and periodic variation and the covariance between the two time periods depends only on its lags and it is time invariant. The most popular method of stationarity detection is graphical analysis and the unit root test.

A stationary time series is easy to predict because one can estimate its future properties since mean and variance will be the same as they were in the past. There are two types of stationary series (a) Strong stationary (b) weak stationary

A strict or strong stationary process is a stochastic process whose unconditional joint probability distribution does not change when shifted in time. Consequently, parameters such as mean and variance, if they are present, also do not change over time. whereas, if a time series has constant mean and variance throughout the times then it is known as weak stationary process.

Stationarizing a time series through differencing (where required) is an important part of the process of fitting a time series model, e.g, ARIMA. The best way to check the stationarity is graphical analysis and unit root tests. The graphical analysis includes Autocorrelation function (ACF) and Partial Autocorrelation function (PACF). Whereas the unit root test include Augmented Dickey Fuller (ADF) test and Phillips Perron (PP) test which will be discussed later. Generally, non-stationarity in the time series data can be due to many reasons. A non-stationarity time series is one for which the parameters are functions of time, i.e., its mean, variance and so on, change over time. A time series is said to be stationary when it contains no growth or decline otherwise will be called non-stationary. When a time series is non-stationary, autocorrelations will dominate the pattern. To model the non-trend pattern in the series, trends must be removed before further analysis. The most important technique is to carry out consecutive differencing of the series to achieve stationarity and then fit a suitable model to differenced data. In fact many time series can be made stationary by replacing the original data points with their first difference; i.e., between successive observations.

3.1.1 Autocorrelation Function

Autocorrelation function (ACF) is often very useful to give the explanation of stochastic process. It describes about the correlation which exist between the data point in the series. It tells about how much correlation exist between the neighbouring data points in the series Y_t . Suppose $Y_1, Y_2, \dots, Y_t$ are measurement at time $1, 2, \dots, t$, the lag k autocorrelation function is defined as

$$\rho_k = \frac{\sum_{i=1}^{t-k} (Y_i - \bar{Y})(Y_{i+k} - \bar{Y})}{\sum_{i=1}^t (Y_i - \bar{Y})^2} \quad (3.1)$$

or more generally

$$\rho_k = \frac{E(Y_t - E(Y_t))(Y_{t+k} - E(Y_{t+k}))}{\sqrt{E(Y_t - E(Y_t))^2 E(Y_{t+k} - E(Y_{t+k}))^2}} \quad (3.2)$$

where Y_t and Y_{t+k} are time series variables and ρ_k is a correlation coefficients, instead of correlation between two different variables, the correlation between two values of the same variable at times X_i and X_{i+k} . The range of ACF is -1 to +1. A good autocorrelation of +1 represents perfect positive correlation, while a value of -1 represents perfect negative correlation. Zero indicates no correlation.

For stationary process, the variance at time t and $t+k$ is same, thus the denominator in

equation 3.2 is just the variance of the stochastic process.

$$\rho_k = \frac{E(Y_t - E(Y_t))(Y_{t+k} - E(Y_{t+k}))}{\sigma_Y^2} \quad (3.3)$$

A plot of ρ_k against k is known as the autocorrelogram or autocorrelation function and is often good guide to the properties of the series.

3.1.2 Partial Autocorrelation Function

Partial Autocorrelation function (PACF) measures the correlation between an observation k periods ago and the current observation, after controlling for observations at intermediate (i.e. all lags $< k$). In general, a partial auto correlation is a conditional correlation. It is the correlation between two variables under the assumption that we know and take into account the values of some other set of variables. For instance, consider a regression context in which Y is a response variable and x_1 , x_2 , and x_3 are predictor variables. The partial correlation between Y and x_3 is the correlation between the variables determined taking into account how both Y and x_3 are related to x_1 and x_2 . In regression, this partial correlation could be found by correlating the residuals from two different regressions:

- Regression in which we predict Y from x_1 and x_2
- regression in which we predict x_3 from x_1 and x_2

Basically, we correlate the parts of Y and x_3 that are not predicted by x_1 and x_2 . More formally, we can define the partial correlation just described as

$$\rho_k = \frac{Covariance(Y, x_3 | x_1, x_2)}{\sqrt{Variance(Y | x_1, x_2) Variance(x_3 | x_1, x_2)}} \quad (3.4)$$

3.2 Stationarity

In practice a researcher is always interested to know the theoretical mean, variance and autocorrelation of a series which are unknown. When a time series posses of a constant mean, a constant variance over time and the autocorrelation function that depends only on the length of the expressed lags then such a series is called stationary or covariance stationary. In covariance stationary series mean and auto-covariances of a series are unaffected by change in time origin. Formally a time series is covariance stationary if for all t

and t-s,

$$\begin{array}{ll} E(Y_t)=E(Y_{t-s})=\mu & \text{constant \& independent of time} \\ \text{var}(Y_t)=\text{var}(Y_{t-s})=\sigma^2 < \infty & \text{constant \& independent of time} \\ \text{cov}(Y_t - Y_{t-s})=\gamma_s & \text{autocovariances} \end{array}$$

Here $t=1,2,3,\dots,T$, and s =number of lags

If a series do not possess the above properties then the series is called non-stationary. There are several methods to check whether series possesses the properties of stationarity or not. Analyzing graphs of the series is one of them which provides a rough idea about the stationarity. In this method we mostly use the following two techniques ([Shah and Waleed, 2010](#)).

- The most common technique is visual analyzing of a time series by observing the line graph for the data. A plot of stationary series is always fluctuates around its mean. A series with definite upward or downward trend with the passage of time is often representing a non stationary series.
- The second technique to get a rough idea about stationarity is to observe correlogram of autocorrelation function. For a stationary time series, the ACF tend to zero rather quickly at initial lags values. While for a non stationary series the ACFs are suffered from linear decline.

3.3 Unit Root Test

Unit root test help us in investigating whether a given series contains a trend and the nature of trend that is deterministic or stochastic. When any series contains a unit root, it tells us about the non stationarity of the series.

There are several methods to test for unit root in the series. In our analysis we use Augmented Dickey Fuller (ADF) and Phillip Perren's (PP) tests to check unit root in our data and to decide about the presence or absence of stationarity in the data.

3.3.1 Augmented Dickey Fuller test

Augmented Dickey Fuller test (ADF) is a test for a unit root in a time series. This test is an extension of the Dickey-Fuller (DF) test for a larger and more complicated set of time series model. This test provides a formal test for non stationarity in time series data. ADF test is used to test the presence of unit root in the coefficient of a lagged variables in a time series model. If the coefficient of a lagged variable in a time series model shows a value of one, then this is an indication of unit root in the series.

To test the presence of a unit root in a time series, [Dickey and Fuller \(1979\)](#) considered the estimation of the parameter γ from the following models.

$$\delta Y_t = \gamma Y_{t-1} + \alpha_2 \delta Y_{t-1} + \dots + \epsilon_t \quad \text{Random walk}$$

$$\delta Y_t = \alpha_o + \gamma Y_{t-1} + \alpha_2 \delta Y_{t-1} + \dots + \epsilon_t, \quad \text{Random walk with drift}$$

$$Y_t = \alpha_o + \beta t + \gamma Y_{t-1} + \alpha_2 \delta Y_{t-1} + \dots + \epsilon_t, \quad \text{Random walk with drift and time trend}$$

it is assumed that $Y_o = 0$ and $\epsilon_t \sim iid(0, \sigma^2)$

We test the null hypothesis $H_o: \gamma = 0$, for the above three models.

Decision rule:

- When ADF test statistic value $>$ test critical value then, accept the null hypothesis.
Unit root exist in the series. The series is not stationary.
- When ADF test statistic value $<$ test critical value then, reject the null hypothesis.
Unit root do not exist in the series. The series is stationarity.

When null hypothesis of unit root is accepted then we difference the data before running a appropriate model for the series. If the null hypothesis is rejected, the series is stationary and can be used without differencing.

3.3.2 Phillips-Perron Test

Phillip-Perron develops a formal procedure to test for unit root in the presence of a structural change at time period $t = \tau + 1$. [Phillips and Perron \(1988\)](#) proposed non-parametric transformations of the τ statistics from the original DF regressions such that under the unit root null, the transformed statistic (the “z” statistic) have DF distributions.

Model under the null hypothesis

$$H_o : Y_t = \alpha_o + \alpha_1 Y_{t-1} + \mu_1 D_p + \epsilon_t$$

Here D_p represent a dummy variable such that $D_p = 1$ if $t = \tau + 1$ and zero otherwise. Model under the alternative hypothesis

$$H_1 : Y_t = \alpha_o + \alpha_2 Y_{t-1} + \mu_2 D_L + \epsilon_t$$

Here D_L is a level dummy variable such that $D_L = 1$ if $t > \tau$ and zero otherwise.

Decision Rule:

Accept the null hypothesis of the unit root if the calculated value of the t-statistics is greater than the critical values.

3.4 Autoregressive Process

Autoregressive (AR) rotates around regressing the variable of interest on its earlier terms. For example, AR (p) or, equivalently, ARIMA(p,0,0) is given by

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2}, \dots, \phi_p Y_{t-p} + \mu_t \quad (3.5)$$

Where $\phi_1, \phi_2, \dots, \phi_p$ are the parameters of the model and μ_t is white noise. In AR modeling, observations of previous time period is treated as predictor for current time period. The AR model comprise of only one laged value is called AR(1). If the current value depends on two previous values, such an AR model is of order two. Similarly, an AR model where the current value linearly depends on p past values is called AR(p). The ACF of AR model decreases slowly represent decreasing relationship of present with future with respect to time. The current value correlates with all previous values and have long memory.

3.5 Moving Average Process

Moving Average (MA) states that the output variable depends on the present and past value of a stochastic (disturbance) term. The MA (q) process is represented as

$$Y_t = c + \theta_1 Z_{t-1} + \theta_2 Z_{t-2} + \dots + \theta_q Z_{t-q} + Z_t \quad (3.6)$$

Where q determine the number of terms included in the model. θ_i (i=1,2,3,...,q) are the parameters of the model. It can also be represented by ARIMA (0,0,q). MA model is used for short run forecasting. If the current value regressed on error term of last past values then it is Moving Average of order one, denoted by MA (1). If current value

regressed on error term of past two values then it is moving average of order two, denoted by MA (2). If current value regressed on error term of previous q period then it is moving average of order q and is denoted by MA (q). For MA models, the ACF will dampen exponentially and the PACF will be used to identify the order (q) of the MA model. If we have no significant spike at lag 0 on the PACF, then we have an MA model of the order 0, i.e., MA (0). If we have significant spikes at lag 1,2, and 3 of the PACF, then we may have an AR model of the order 3, i.e., AR(3).

3.6 Autoregressive moving average Model

Autoregressive moving average (ARMA) model consider the AR and MA component to model the time series. If integration is included, the ARMA model is written as Autoregressive integrated moving average (ARIMA) model. The ARMA (p,q) can be written as

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \theta_1 Z_{t-1} + \theta_2 Z_{t-2} + \dots + \theta_q Z_{t-q} + Z_t \quad (3.7)$$

The equation could also be written as :

$$Y_t - \phi_1 Y_{t-1} - \dots - \phi_p Y_{t-p} = Z_t + \theta_1 Z_{t-1} + \dots + \theta_q Z_{t-q} \quad (3.8)$$

where $Z_t \sim WN(0, \sigma^2)$ and ϕ_p and θ_q are the coefficients of AR and MA respectively. it is convenient to use more concise form of (3.8)

$$\phi(B)Y_t = \theta(B)Z_t \quad (3.9)$$

where $\phi(\cdot)$ and $\theta(\cdot)$ are the p^{th} and q^{th} degree polynomials such that $\phi_z = 1 - \phi_1 z - \dots - \phi_p z^p$ and $\theta_z = 1 + \theta_1 z + \dots + \theta_q z^q$ where B is the backward shift operator ($B^j X_t = X_{t-j}$, $B^j Z_t = Z_{t-j}$).

Y_t is the values of the time series observation at time t, Z_t is a series of random shocks which are assumed to be independently normally distributed with constant mean say zero, variance and uncorrelated with each other.

ARMA models consist of two components: lagged values of the variable of interest (the AR component) and lagged values of the error term (the MA component). ARMA process

can be used to model a large class of stationary time series as long as the appropriate order of p and q are specified [Makridakis and Hibon \(1997\)](#). Such model states that the current value of some series Y_t depends linearly on its own previous values plus a combination of the current and previous values of an error term.

3.7 Box and Jenkins Methodology

Basically the fitting of ARIMA (p,d,q) model to time series consists of the following steps

(a) Model Identification (b) Parameter Estimation (c) Adequacy of Model (d) Forecasting

3.7.1 Model Identification (or Specification)

The identification of the ARMA (p,q) model is specified on the basis of correlograms computed for the stationary time series. The significance of ACF and PACF are checked by comparing with two standard error bound. The general theoretical patterns of ACF and PACF are given in following table

	AR(p)	MA (q)	ARMA (p,q)
ACF	tails off	cuts off after lag q	tails off
PACF	cuts off after lag p	tails off	tails off

3.7.2 Estimation

The parameters of ARMA(p,q) model are estimated using different statistical techniques such as if an AR (p) process is identified then the least squared (LS) or maximum likelihood (ML) can be used to estimate the parameter. Whereas, for MA (q) process, maximum likelihood estimation (MLE) may be used. Similarly if ARIMA model is identified then parameter can be estimated by using least square or maximum likelihood estimation.

3.7.3 Model Diagnostic

To select most appropriate model for forecasting, different error summary measures are used. Root mean square error (RMSE), Mean absolute error (MAE), Mean absolute percentage error (MAPE) are used to check the performances of the model in this study. Root mean square error is the square root of Mean Square Error (MSE). It is the most sensitive

error. It squares the errors which assigns more weight to large errors. Mean absolute error is defined as the average of the absolute errors. It can be compared to other errors if they have the same units. It is less sensitive to large errors and hence slightly smaller than RMSE. Mean Absolute percentage error is defined as the average of relative absolute error for each sample point. It gives large value even undented values when the observed value is close to zero. It imposes penalty on negative values. The detailed description of all these error summary measures are given at the end of this chapter.

3.7.4 Forecasting

Once an adequate ARMA(p,q), AR(p), MA(q) models are specified the next step is the forecasting. The computation of the forecast can be done recursively using the estimated models. This involves first computing a forecast one period ahead, using this forecast to compute a forecast two periods ahead, and continuing until the desired period forecast has been reached. The forecasts are made by applying models of each series.

3.8 Volatility Models

This section contains an introduction and estimation of volatility models which includes ARCH, GARCH and EGARCH. According to [Charles and Darné \(2005\)](#), the differences in volatility models are due to the way variance develops gradually over time. The ARCH/GARCH models are selected because of the restriction of the ordinary least square (OLS). A major assumption behind ordinary least square is that the error terms have a constant variance (a property of OLS commonly known as homoscedastic). If this condition is violated, then the ordinary least square estimates will not have minimum variance even though the coefficients will still be unbiased estimates. The ARCH/GARCH models handle this problem by modeling volatility which depends on previous volatility in the regression equation and correcting the deficiencies of least square model. To capture volatility clustering and make it mean reverting, ARCH model is preferred over the other model which fail to capture these important factors of time series data.

3.8.1 ARCH Model

Autoregressive conditional heteroskedasticity (ARCH) model is a statistical model for time series data that describes the variance of the current error term or innovation as a function of the actual sizes of the previous time periods error terms, often the variance is related to the squares of the previous innovations. ARCH models are commonly employed in

modeling financial time series that exhibit time-varying volatility and volatility clustering, i.e. periods of swings interspersed with periods of relative calm. ARCH-type models are sometimes considered to be in the family of stochastic volatility models.

To model a time series using an ARCH process, let ϵ_t denote the residual series of an AR process. These ϵ_t are split into a stochastic piece z_t and a time-dependent standard deviation σ_t characterizing the typical size of the terms so that $\epsilon_t = \sigma_t z_t$, where z_t is strong white noise process. The series σ_t^2 is modeled by

$$\sigma_t^2 = \alpha_o + \alpha_1 \epsilon_{t-1}^2 + \dots + \alpha_q \epsilon_{t-q}^2 = \alpha_o + \sum_{i=1}^q \alpha_i \epsilon_{t-i}^2$$

The ARCH model have been widely applied to the study of stock return rate. The property of “uncertainty or unpredictability” in determining the behavior of various economic variables, force economists to replace the non-conditional mean as in ARMA models for the conditional mean as a contribution towards improving forecast. An ARCH process is a procedure that consist of past variances in the explanation of future variances. The Autoregressive property explain a comeback response process that in corporates past observations into the present while Conditionality indicates the existence of a dependence on the observations of the immediate past and heteroscedasticity means time-varying variance.

Inspite of the significance of the ARCH model for many financial time series, a relatively long lag length in the variance equation with the problem of estimation of parameters subject to inequality limitation is often called for to capture the long memory typical of financial data. The order of p can be obtained by using the AIC or the Sample Autocorrelation function of the squared returns where the sample PACF is not significantly different from zero for lags greater than p in the ARCH model. The ARCH model requires a high order to accurately model conditional variance.

3.8.2 GARCH Model

GARCH model is the generalization of ARCH model introduced by [Bollerslev \(1986\)](#). Conditional variance depends on the weighted average of past squared residuals and it assigns a lesser weight to past squared residuals which does not go completely to zero in GARCH model. The most commonly used GARCH model states that the appropriate predictor of the variance in the next period is a weighted average of the long run average variance. A time series ϵ_t is given at each instance by:

$$\epsilon_t = \sigma_t \omega_t$$

Where ω_t is white noise, with zero mean and unit variance, and σ_t^2 is given by:

$$\sigma_t^2 = \alpha_o + \sum_{i=1}^q \alpha_i \epsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2 \quad (3.10)$$

Where α_i and β_j are parameters of the model. We say that ϵ_t is a generalised autoregressive conditional heteroskedastic model of order p,q, denoted by GARCH(p,q). Zivot (2009) suggested that a GRACH (1,1) model with only three parameters in the conditional variance equation is enough to obtain a good fit for financial time series. GARCH model cannot be used to explain the leverage effects and does not allow for direct feedback between the conditional variance and the conditional mean. An asymmetric GARCH model called EGARCH model developed by Nelson (1991) to measure the volatility of returns of stock market considered for these reasons.

3.8.3 EGARCH Model

The EGRACH model has the unique ability of volatility persistence, mean revision as well as asymmetrical effect. It has the ability to split the effect of positive and negative shocks on volatility, makes it different from the GARCH model. The model does not impose any limitation on the parameters to ensure that the variance is nonnegative.

Standard GARCH models assume that positive and negative error terms have a symmetric effect on the volatility. In other words, good and bad news have the same effect on the volatility in the model. In practice this assumption is frequently violated, in particular by stock return, in that the volatility increases more after bad news than after good news.

This so called Leverage Effect defined by Black (1976), who noted that:

“a drop in the value of the firm will cause a negative return on its stock, and will usually increase the leverage of the stock. This rise in the debt-equity ratio will surely mean a rise in the volatility of the stock”.

Let Z_t denotes a series of i.i.d standardized random variables with expectation 0 and variance 1. The general exponential EGARCH model is Nelson (1991)

$$\log \sigma_t^2 = \omega_t + \sum_{k=1}^{\infty} \beta_k g(Z_{t-k}) \quad (3.11)$$

Where ω_t , β_k are deterministic coefficients and

$$g(Z_t) = \theta Z_t + \gamma(|Z_t| - E|Z_t|) \quad (3.12)$$

it can be directly seen that $E[g(Z_t)] = 0$

The EGARCH models shows some differences from the standard GARCH model:

- Volatility of the EGARCH model, which is measured by the conditional variance σ_t^2 , is an explicit multiplicative function of lagged innovations. On the contrary, volatility of the standard GARCH model is an additive function of the lagged error terms ϵ_t , which causes a complicated functional dependency on the innovations
- Volatility can react asymmetrical to the good and bad news
- For the general distributions of z_t the parameter restrictions for strong and covariance-stationarity coincide.

EGARCH models benefit from no parameter restrictions, thus the possible instabilities of optimization routines are reduced.

Let $\omega_t = \omega = 0$ and $\sum_{k=1}^{\infty} \beta_k^2 < \infty$. Then σ_t^2 is strictly stationary and ergodic with reference [Nelson \(1991\)](#). Furthermore, under these conditions the unconditional variance exists when Z_t has a generalized error distribution (GED) with parameter $\gamma > 1$, which is determines the thickness of thee tails, [Nelson \(1991\)](#). The GED is leptokurtic when $\gamma < 2$. The normal distribution is a special case of GED ($\gamma = 2$). [Nelson \(1991\)](#) gives additional complicated formulas of the unconditional moments. One problem is that under other leptokurtic distributions such as Student-t, the unconditional variance does not exist. The reason is that exponential growth of the conditional variance changes with the level of the shocks, which leads to the explosion of the unconditional variance when the probability for extreme shocks is sufficiently large. Therefore the existence of the unconditional moments depends on the choice of the distribution of the innovations, which is an undesirable property of the EGARCH models. In empirical studies it has been found that EGARCH often over-weighs the effect of larger shocks on volatility and thus results in poorer fits than standard GARCH models ([Engle and Ng, 1993](#)).

3.9 Error Summary Measures

To assess the performance of our predictive models, two well known measures for selecting a model summary measures namely, Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE) are used. It is worth mentioning that AIC and BIC are also used for the selection of models, but the importance of RMSE and MAPE is more in time series context.

If we define $e_t = (Y_t - \hat{Y}_t)$ as the one-step-ahead prediction error for day t , where Y_t and $\hat{Y}_t$ are the observed and forecasted values, the descriptive indicators can be written as

$$\begin{aligned}\text{MAPE} &= \text{mean}(|e_{t,j}|/Y_t) \times 100 \\ \text{RMSE} &= \sqrt{\text{mean}(e_{t,j})^2}\end{aligned}$$

Chapter 4

Predictive Models

This chapter briefly describes the nature of data, result of unit root tests and performance of volatility models in capturing volatility.

4.1 Collection of Data

For the current study, data is collected from Pakistan Stock Exchange (PSX) official website. The collected data covers a period of 5 years i.e. from 1st January 2013 to 30th December 2017. The macroeconomic variable "Daily closing prices of stock market" is analyzed for study. Weekends, Public holidays and unusual holidays due to unexpected situation in the country were removed from the data.

For some days there were two indexes i.e. closing index of previous day and another index before opening of other day market. This was usually high index compares to the last day closing index. The reason is that some companies offers to sell their shares before opening of stock market and if expected returns of that particular company is good, then people usually buy those shares and index increases.

It is also observed that the stock returns are highly unpredictable and any uncertain condition effects the market instantaneously. A smooth upward trending market at noon results in crash of market in evening is also part of the history of stock market.

Figure 4.1 shows the graphical representation of data

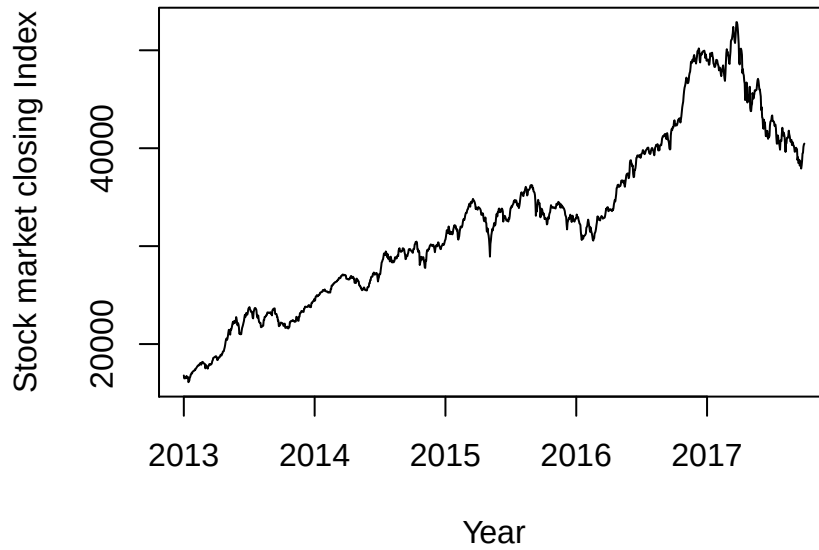


Figure 4.1: PSX 100 index series for the period 1st Jan,2013 to 31st Dec,2017

4.2 Preliminary Analysis

Preliminary analysis involves the editing of data in order to make it useful for further exploratory analysis. It involves fundamental attributes of the data and gives the summarized results like descriptive statistics, some basic plots etc.

Figure 4.1 shows the time series plot of closing PSX 100 index. It can be observe from the plot that series is non-stationary. The series shows random behavior as the trend is upward till 2015, a mix behavior in 2015-2016 and a dramatic upward change in 2017. In this year, the series is reached to its highest ever position in its history. From the end of 2017, the series showed a decreasing trend. This decreasing trend was due to the political uncertainty as the Supreme court of Pakistan disqualify elected Prime Minister, due to which political vacuum, instability and uncertainty increased, that directly affect the stock market. Investors were reluctant in investing in the market and market start declining continuously. This is a practical example of the effect of political instability on stock market returns which is discussed in introductory chapter.

As mentioned above the series is not stationary and to convert non-stationary series into stationary series, we use differencing method by taking a 1 lag difference of the original series. In financial time series, often the series is transformed by log and then the differencing is performed. This is because financial time series is usually exposed to random shocks,

and thus log transformation can smooth out (linearized) the series whereas, differencing will help in de-trending.

The differenced series is plotted in Figure 4.2.

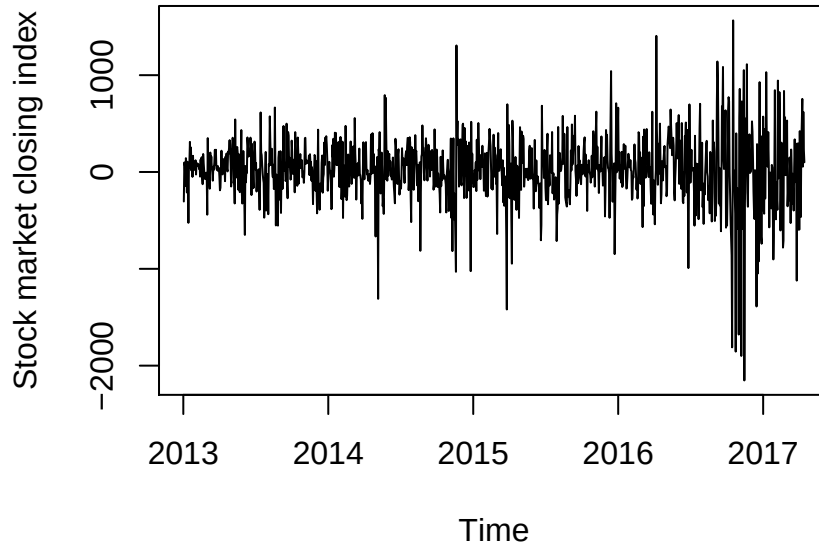


Figure 4.2: 1st order differenced series

From the plot we can see that the trend is vanished and the series is more look like as a stationary series. Hence can be used for modeling purposes. The descriptive statistics for stock market data is given in table 4.1.

Variable	PSX time series
Minimum	16108
1st quartile	25854
Median	32506
Mean	32632
3rd quartile	39465
Maximum	52876

Table 4.1: Descriptive statistics for stock market data

4.3 ACF and PACF of Data

We will present ACF and PACF of original data in this section and also ACf and PACF of differenced data and find the changes after differencing of data.

4.3.1 ACF and PACF of original data

The ACF for stock market in figure 4.3 show geometrical decline which indicates the lack of stationarity in the data.

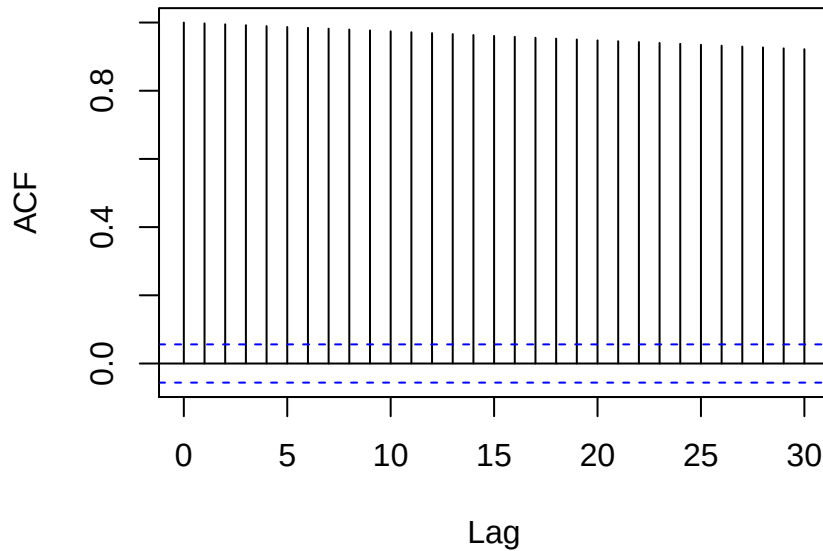


Figure 4.3: ACF of original series

The spikes are decreasing gradually and this pattern indicates that AR model will be appropriate for this series. But we need to remember that our data is non-stationary and we cannot apply model on this. So we need to check ACF and PACF of differenced data also.

Whereas, PACF in figure 4.4 showed only one significant spike and MA model is suggested for this.

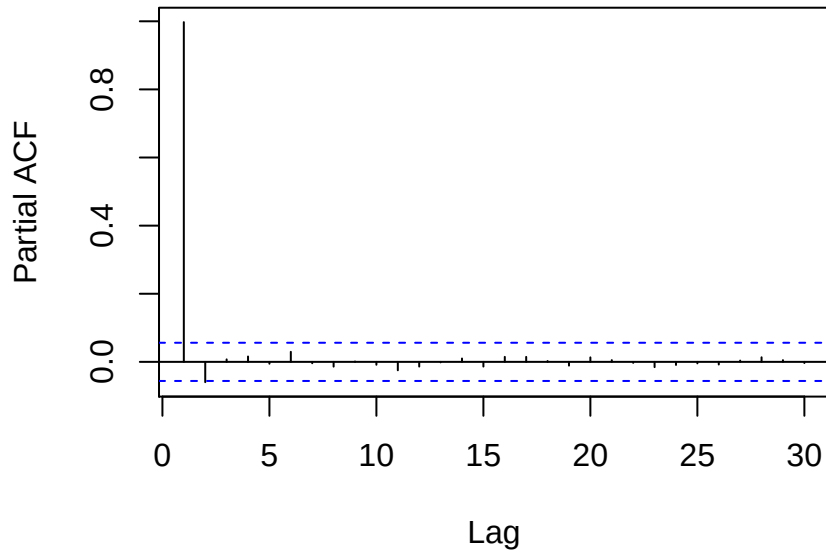


Figure 4.4: PACF of original series

4.3.2 ACF and PACF of Differenced Data

As we mentioned that series becomes stationary after first order differencing. So the ACF and PACF will also change accordingly.

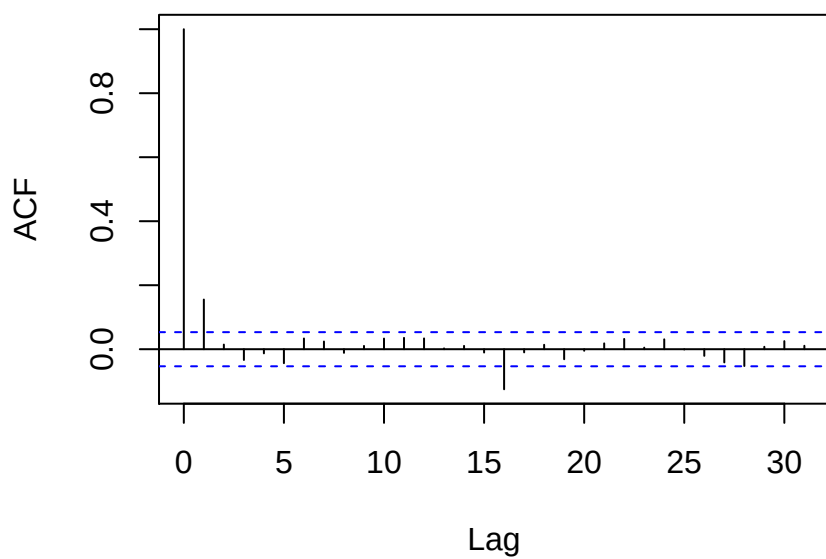


Figure 4.5: ACF of 1st order differenced series

Figure 1.5 shows the ACF of 1st order differenced data. There are 2 significant spikes.

And this trend shows that MA (2) model will be appropriate for this study.

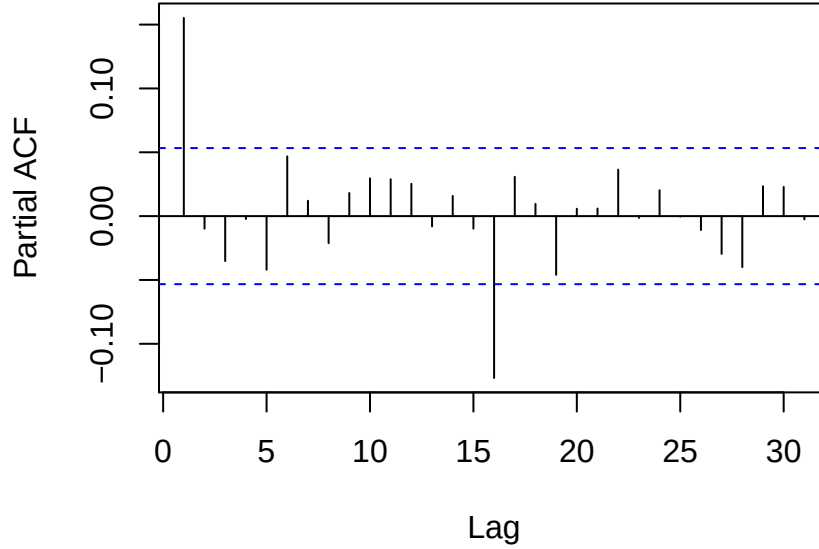


Figure 4.6: PACF of 1st order differenced series

Figure 4.6 shows that PACF of differenced series is cutting after 1-lag and this shows that AR(1) will be appropriate model for this study.

4.4 Application of Data

In this section we perform different unit root tests and cointegration for stock market returns. In our first analysis we perform ADF and PP tests for the stock market series to check stationarity in the data.

4.4.1 ADF Test Results

The null and alternative hypothesis of ADF test is given as

H_o : There is non-stationarity in the series

H_1 : There is stationarity in the series

Test	Test statistics	P value	Null hypothesis	Results
ADF	-1.6293	0.7352	Accept	Non-stationary

Table 4.2: ADF test result before differencing of time series

As p-value of stock returns shown in Table 4.2 is greater than $\alpha=0.05$ which means the null hypothesis cannot be rejected and may conclude the given series is non-stationarity (not have zero mean) and differencing is applied to make it stationarity.

Now we apply ADF test on differenced series and it gives us the following results.

Test	Test statistics	P value	Null hypothesis	Results
ADF	-23.156	0.01	Reject	stationary

Table 4.3: ADF test result after differencing of time series

As now P value is less than 0.05 and we can reject our null hypothesis and conclude that the series is stationary.

4.4.2 PP Test Results

PP test is also used to check stationarity in the series. Table 1.2 indicating that series is non-stationary and the results of PP test after taking first difference are given in Table 4.4

Test	Test statistics	P value	Null hypothesis	Results
PP	-974.7	0.01	Reject	stationary

Table 4.4: PP test result after differencing of time series

4.5 Performances of Models

In this section we are going to discuss the performances of different models. And on the basis of their performances we will choose appropriate model for our study.

4.5.1 ARIMA Model

Autoregressive Integrated Moving average (ARIMA) are usually not recommended by researchers for highly volatile data, as these models are not well capable of capturing sudden changes. We applied these models on our data and compare their performance with other volatility models.

Table 4.5 is showing MAPE and RMSE values of different ARIMA models

Model	MAPE	RMSE
ARIMA (1,1,1)	0.62873	337.12308
ARIMA (1,0,0)	0.63876	337.08359
ARIMA (1,1,2)	0.62518	336.03516
ARIMA (2,1,2)	0.62910	337.71860
ARIMA (2,0,0)	0.62871	337.16064

Table 4.5: MAPE and RMSE values of different ARIMA models

MAPE is choose as the decision variable in our study. So a model having lowest MAPE value will be consider for PSX 100 study. From Table 4.5 we can observe that ARIMA (1,1,2) has the lowest MAPE and RMSE error. So ARIMA (1,1,2) is chosen for PSX 100 study and applied to the series.

4.5.2 Forecasted Results of ARIMA (1,1,2)

After selection of appropriate ARIMA model, we applied ARIMA (1,1,2) on our PSX 100 series and made forecasting for future. Figure 4.7 is showing the forecasting performance of ARIMA (1,1,2).

From the plot we can see that there are some spots at where the observed and forecasted points are at a considerable difference. But there are also some points at where the difference between observed and forecasted is very minimum.

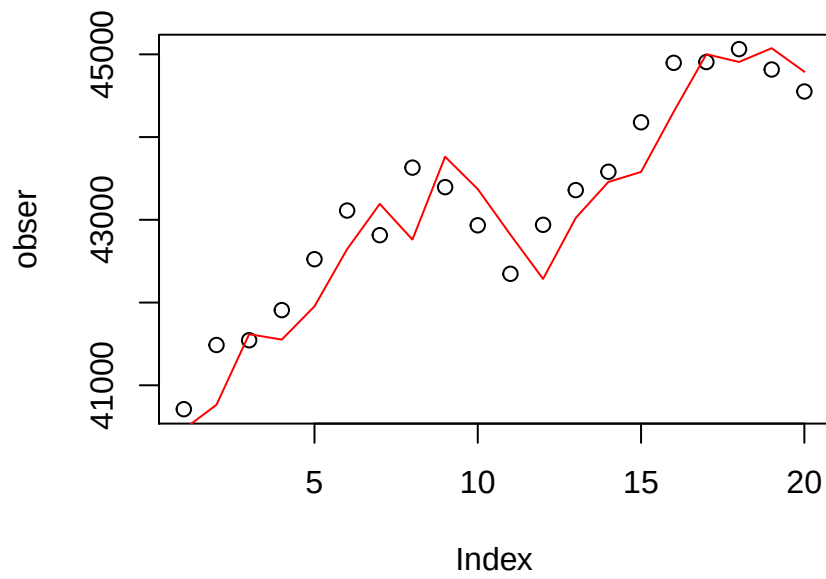


Figure 4.7: Forecasting by ARIMA (1,1,2) model

ACF and PACF of the model

ACF and PACF generated by ARIMA (1,1,2) are also given in figure 4.8 and 4.9 respectively

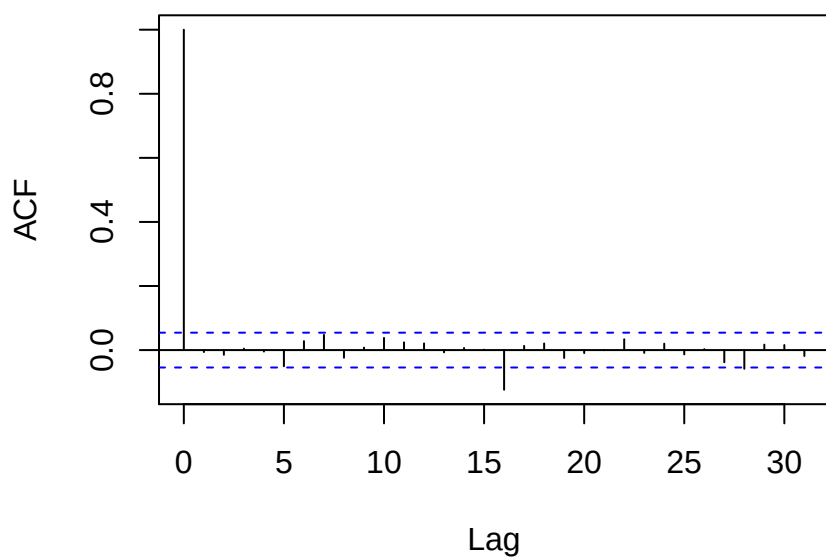


Figure 4.8: ACF generated by ARIMA (1,1,2) model

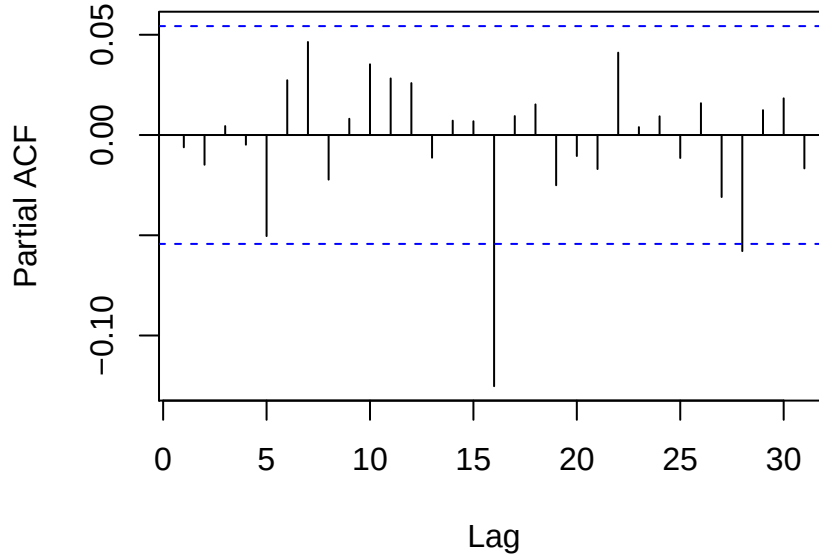


Figure 4.9: PACF generated by ARIMA (1,1,2) model

4.5.3 ARCH Model

Autoregressive conditional heteroskedasticity (ARCH) models are known for their better performance in capturing volatility. Stock markets are highly diversified and reactive to minor indicators. It is not surprising to know that stock markets in developed countries are also directly effected by many macro and micro indicators. So in these situations models like ARCH are recommended to use.

Table 4.6 gives the MAPE and RMSE of different ARCH models applied on our study

Model	MAPE	RMSE
ARCH (1,0)	0.71448	391.11671
ARCH (1,1)	0.62183	338.14514
ARCH (2,1)	0.62902	337.36105
ARCH (3,1)	0.61907	336.01860
ARCH (2,0)	0.62773	336.65381

Table 4.6: MAPE and RMSE values of different ARCH models

From Table 4.6 we can observe that ARCH (3,1) gives lowest value of MAPE and RMSE value. so we choose ARCH (3,1) from ARCH family for our PSX 100 series.

4.5.4 Forecasted Results of ARCH (3,1)

Figure 4.10 is showing forecasted graph of ARCH (3,1). We can observe that ARCH model made better forecast than ARIMA model as these models are good in capturing sudden changes.

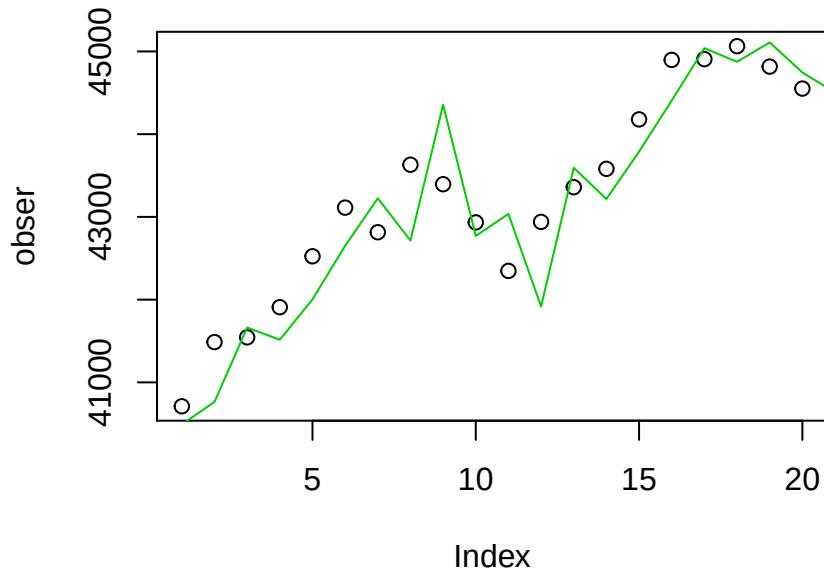


Figure 4.10: Forecasting by ARCH (3,1) model

The ACF and PACF of ARCH (3,1) model are given in figure 4.11 and 4.12 respectively.

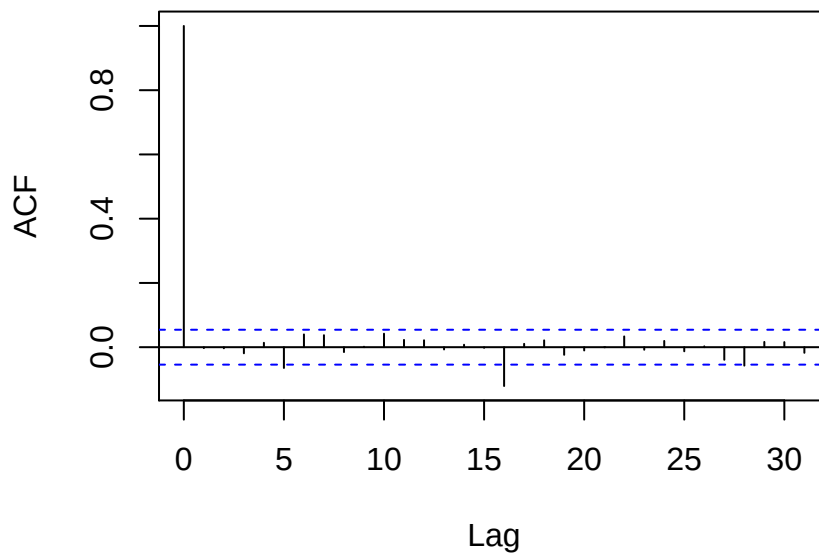


Figure 4.11: ACF generated by ARCH (3,1) model

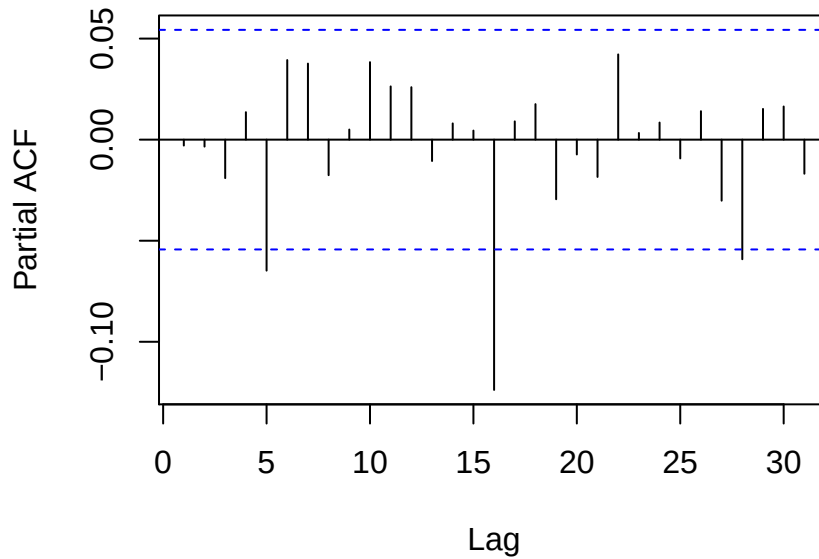


Figure 4.12: PACF generated by ARCH (3,1) model

4.5.5 GARCH Model

GARCH model is the extension of ARCH model and widely used for capturing volatility. The performance of this model is better among other volatility models. Specially GARCH (1,1) is highly recommended for stock market studies.

Model	MAPE	RMSE
GARCH (2,1)	0.63448	340.11194
GARCH (3,2)	0.61105	336.01225
GARCH (1,1)	0.71448	391.11671
GARCH (2,0)	0.63422	339.75678
GARCH (2,3)	0.63168	338.46142

Table 4.7: MAPE and RMSE values of different GARCH models

From Table 4.7 we observe that GARCH (3,2) gives lowest MAPE and RMSE value among all other models.

4.5.6 Forecasted Results of GARCH (3,2)

From figure 4.13 we can observe that how much better forecast we get from GARCH (3,2) model. GARCH (3,2) model predict the volatility to much good extent.

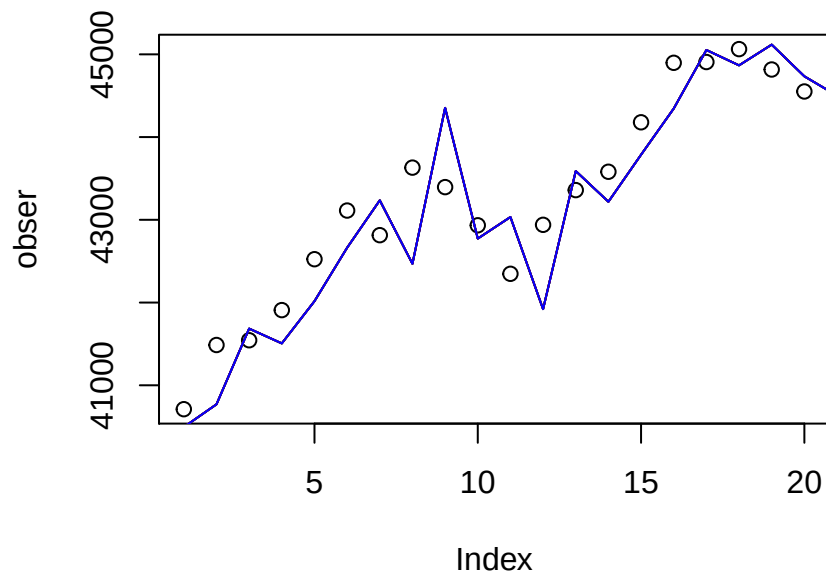


Figure 4.13: Forecasting by GARCH (3,2) model

The ACF and PACF of GARCH (3,2) are given in figure 4.14 and 4.15 respectively:

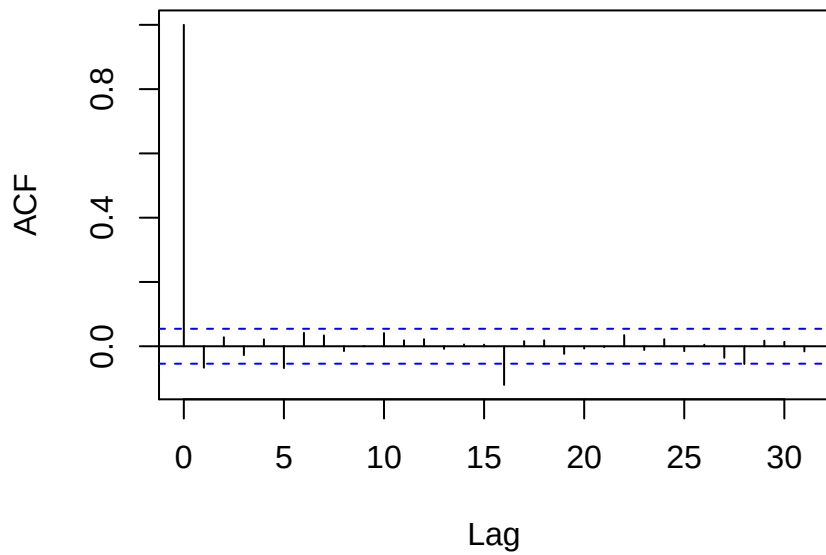


Figure 4.14: ACF generated by GARCH (3,2) model

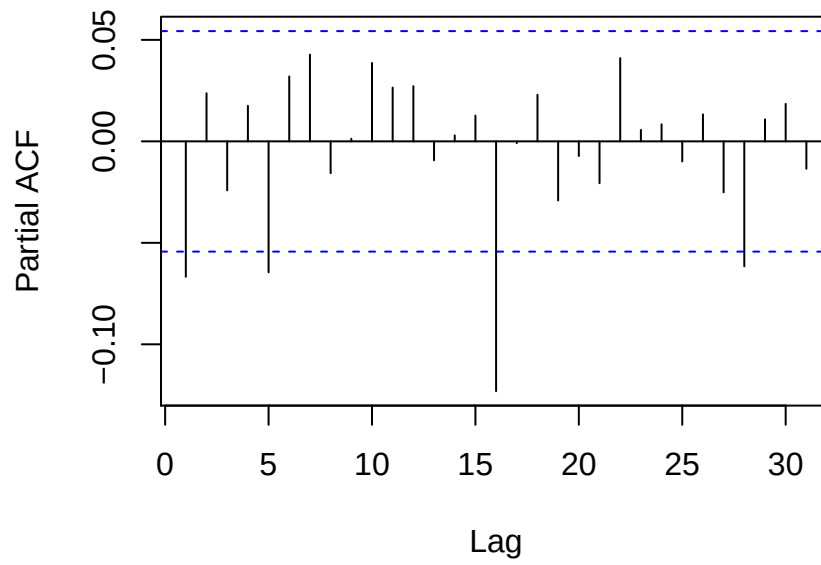


Figure 4.15: PACF generated by GARCH (3,2) model

Chapter 5

Results Analysis and discussion

In this chapter we are going to discuss all the results collectively and compare the performance of different models and choose best model for our study

We observe in previous chapter that GARCH (3,2) gives us the lowest MAPE and RMSE value among all other models. And we also notice that GARCH (3,2) gives us more precise forecasted results. So on the basis of these results we conclude that GARCH (3,2) is appropriate model for our PSX 100 series.

5.1 Summary and future directions

Poor forecasting not only shakes investor confidence but also mark a question on research's skills and ability of forecasting. Human nature is a kind of detective, as he always try to disclose hidden facts and objects, directly or indirectly effecting our lives. Researchers always in a way to find new methodologies and techniques to make better forecasting. But there is always some amount of gap in predicted and actual results and this is what we call error. The ultimate goal of every researcher is to minimize this gap.

Forecasting of Stock market returns is one of challenging task for researchers, as there are several factors effecting stock market returns directly or indirectly. Pakistan Stock market is also one of highly volatile market, so there is always chance of ambiguity and uncertainty. As PSX is a emerging market so it takes no time to effected by external effects like political uncertainty, terrorism etc.

We applied ARIMA, ARCH and GARCH models to capture volatility in PSX and find that ARCH and GARCH models gives comparatively more precise forecast and are recommended for stock market study.

Furure Directions

The following are the some future directions which could be use for future study.

- Further analysis can be done by using some other data and on the basis of those results we can make government vs government comparison.
- Analysis can be done by examining external factors effects e.g., terrorism, inflation, political uncertainty etc.
- Analysis can be done by using extensions of volatility models e.g., TGARCH
- Analysis can be done by using more techniques which are recommended by other researchers

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